

| S.NO | TASK                           | SUBTASK                     | STATUS      | UPDATE DATE | DESCRIPTION                                           | Duration |  |
|------|--------------------------------|-----------------------------|-------------|-------------|-------------------------------------------------------|----------|--|
| 1    | System Refinement & Stability  |                             | In Progress | 10/3/2026   | Lane stability achieved, refining system              | 9 days   |  |
| 1.1  |                                | Lane robustness improvement | Completed   | 11/3/2026   | Improve detection under tough conditions              |          |  |
| 1.2  |                                | Detection stabilization     | Completed   | 12/3/2026   | Reduce YOLO flickering                                |          |  |
| 1.3  |                                | FSM smoothing               | Completed   | 13/03/2026  | Improve transitions                                   |          |  |
| 1.4  |                                | Steering optimization       | Completed   | 14/03/2026  | Reduce oscillations                                   |          |  |
| 1.5  |                                | Continuous run validation   | Completed   | 15/03/2026  | 8 lap validation                                      |          |  |
| 2    | Advanced Behavior Intelligence |                             | Completed   | 16/03/2026  | Parking & overtaking implemented                      | 7 days   |  |
| 2.1  |                                | Dynamic speed control       | Completed   | 17/03/2026  | Speed based on curvature/object                       |          |  |
| 2.2  |                                | Pedestrian response tuning  | Completed   | 18/03/2026  | Improve response timing                               |          |  |
| 2.3  |                                | Priority handling logic     | Completed   | 19/03/2026  | Conflict resolution logic                             |          |  |
| 2.4  |                                | Recovery behavior           | Completed   | 20/03/2026  | Fallback handling                                     |          |  |
| 2.5  |                                | Parking system              | Completed   | 21/03/2026  | Parking implemented                                   |          |  |
| 2.6  |                                | Highway overtaking          | Completed   | 22/03/2026  | Overtaking implemented                                |          |  |
| 3    | Edge Case & Stress Testing     |                             | Completed   | 23/03/2026  | Stress testing phase                                  | 7 days   |  |
| 3.1  |                                | Lighting variation testing  | Completed   | 24/03/2026  | Test performance under different lighting conditions. |          |  |
| 3.2  |                                | Partial lane testing        | Completed   | 25/03/2026  | Handle incomplete or unclear lane markings            |          |  |
| 3.3  |                                | Multi-object testing        | Completed   | 26/03/2026  | Test behavior with multiple objects detected.         |          |  |
| 3.4  |                                | Delay simulation            | Completed   | 27/03/2026  | Evaluate system under delayed input frames.           |          |  |
| 3.5  |                                | Failure fixes               | Completed   | 28/03/2026  | Identify issues and fix root causes.                  |          |  |
| 4    | Performance Optimization       |                             | Completed   | 29/03/2026  | Optimization phase                                    | 4 days   |  |
| 4.1  |                                | Inference optimization      | Completed   | 30/03/2026  | Reduce model inference time.                          |          |  |
| 4.2  |                                | Resource optimization       | Completed   | 31/03/2026  | Optimize CPU and memory usage.                        |          |  |
| 4.3  |                                | Pipeline efficiency         | Completed   | 01/04/2026  | Improve pipeline processing efficiency.               |          |  |
| 4.4  |                                | Frame rate tuning           | Completed   | 02/04/2026  | Optimize frame rate and processing flow.              |          |  |
| 5    | Visualization & Explainability |                             | Completed   | 3/4/2026    | Visualization phase                                   | 3 days   |  |
| 5.1  |                                | Dashboard enhancement       | Completed   | 4/4/2026    | Enhance dashboard with system outputs.                |          |  |
| 5.2  |                                | Decision logging            | Completed   | 4/4/2026    | Log decision-making actions.                          |          |  |
| 5.3  |                                | Telemetry recording         | Completed   | 6/4/2026    | Record system data for analysis.                      |          |  |
| 6    | Final Demo & Video             |                             | Completed   | 07/04/2026  | Demo prep                                             | 2 days   |  |
| 6.1  |                                | Recording                   | Completed   | 11/04/2026  | Record autonomous system performance.                 |          |  |
| 6.2  |                                | Editing                     | Completed   | 11/4/2026   | Edit and finalize demo video.                         |          |  |
| 7    | Documentation                  |                             | Completed   | 10/4/2026   | Docs prep                                             | 2 days   |  |
| 7.1  |                                | Architecture                | Completed   | 12/4/2026   | Create system architecture diagram.                   |          |  |
| 7.2  |                                | Explanation                 | Completed   | 12/04/2026  | Explain complete system workflow.                     |          |  |